HAT-P-36b

R-0742

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_260223` · created 2026-07-28T02:01:48+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461094.9097 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.013
Transit depth (Rp/Rs)^2	1.94 +/- 0.37 [%]
Orbital Inclination (inc)	89.85 +/- 1.51
Airmass coefficient 1 (a1)	1.0856 +/- 0.0088
Airmass coefficient 2 (a2)	-0.078 +/- 0.074
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	#3 - [345, 89]
Optimal Aperture	3.61
Optimal Annulus	9.22
Transit Duration (day)	0.0986 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.013 vs 0.126 (deviation 1.0σ)
Duration fit vs published	0.0986 d vs 0.0925 d (deviation 4.36σ)
Mid-time O–C	1.3 min
Depth SNR	10.7
Residual scatter	0.81 %

Light curve

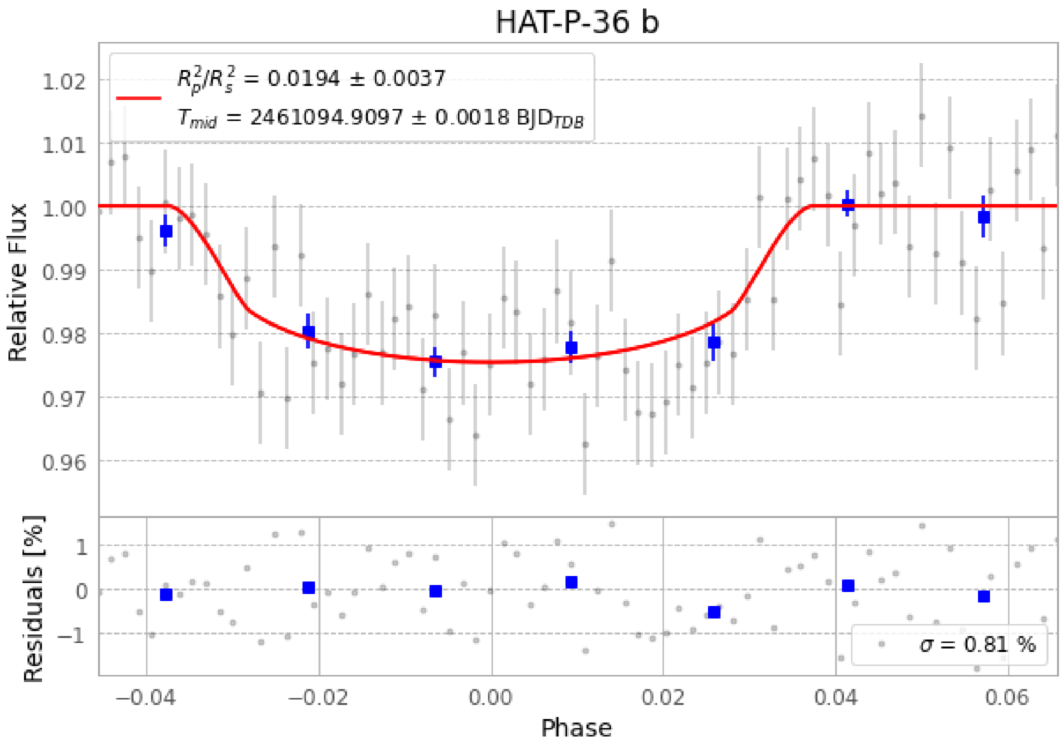

Figure 1 — FinalLightCurve_HAT-P-36 b_2026-02-23.png (SHA-256 6cb0e82c2ec3cd2b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 104], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b8f6951763f2732a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

72 FITS frames (47 MB), `HATP-36260223081516.FITS` → `HATP-36260223114816.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260223.log (8936986090c4…), FinalLightCurve_HAT-P-36 b_2026-02-23.png (6cb0e82c2ec3…), FinalLightCurve_HAT-P-36 b_2026-02-23.csv (f88c538c39d2…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)